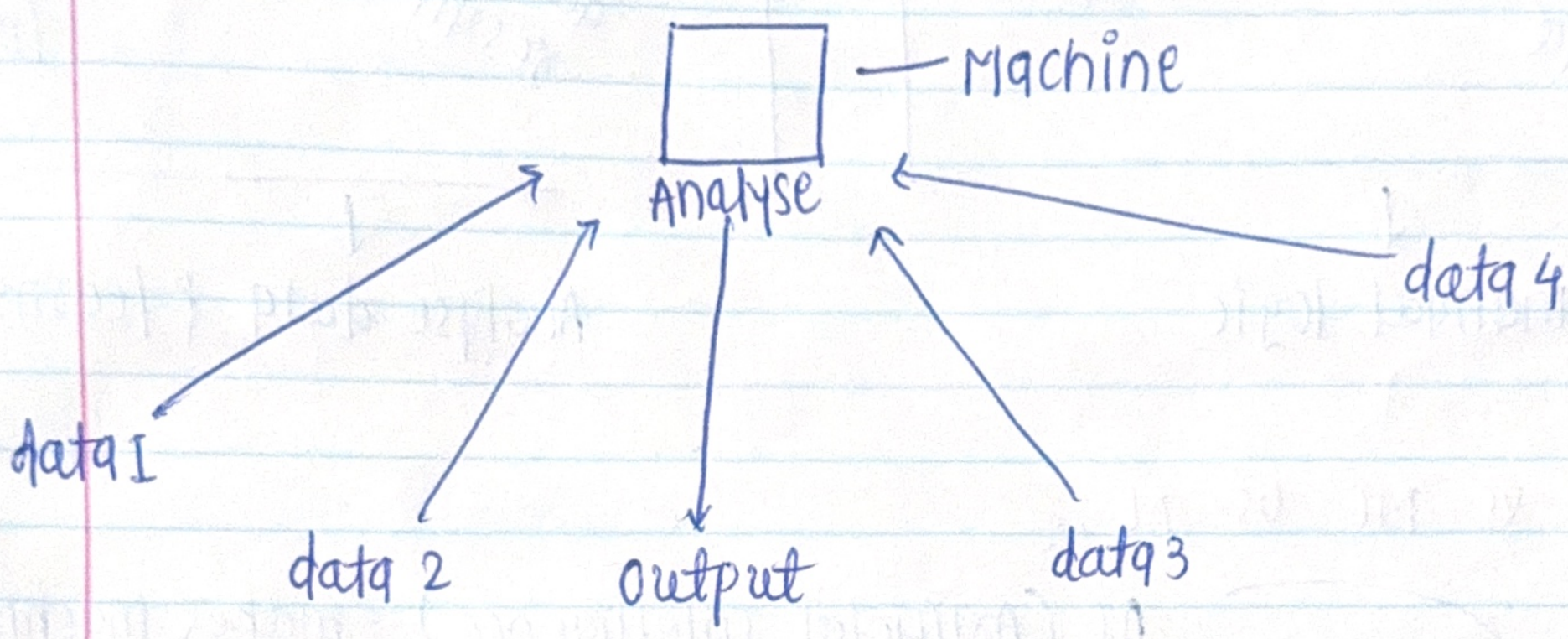
March 12th

# Machine learning

> "Machine learning is all about teaching computers to learn from data - just like we learn from experience."

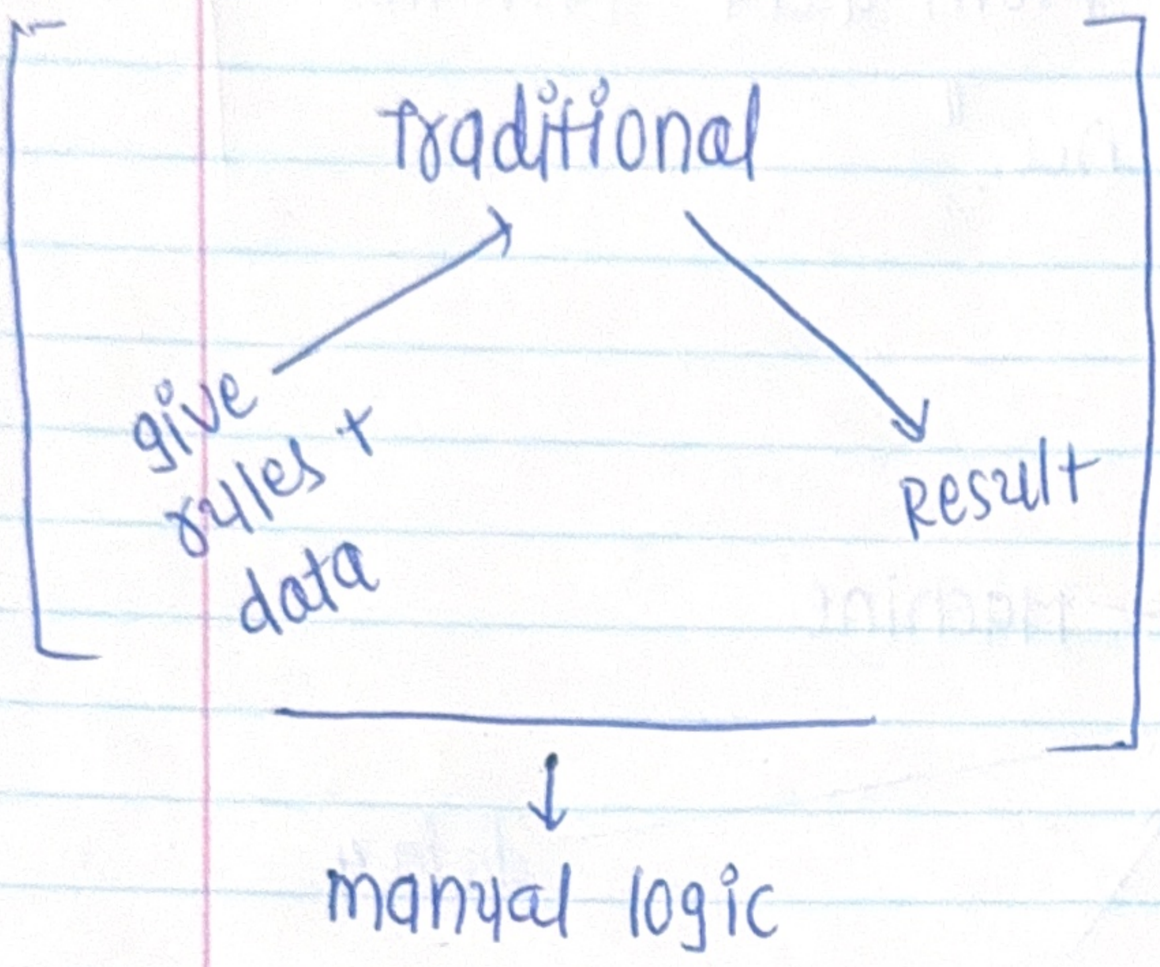
* What is machine learning?

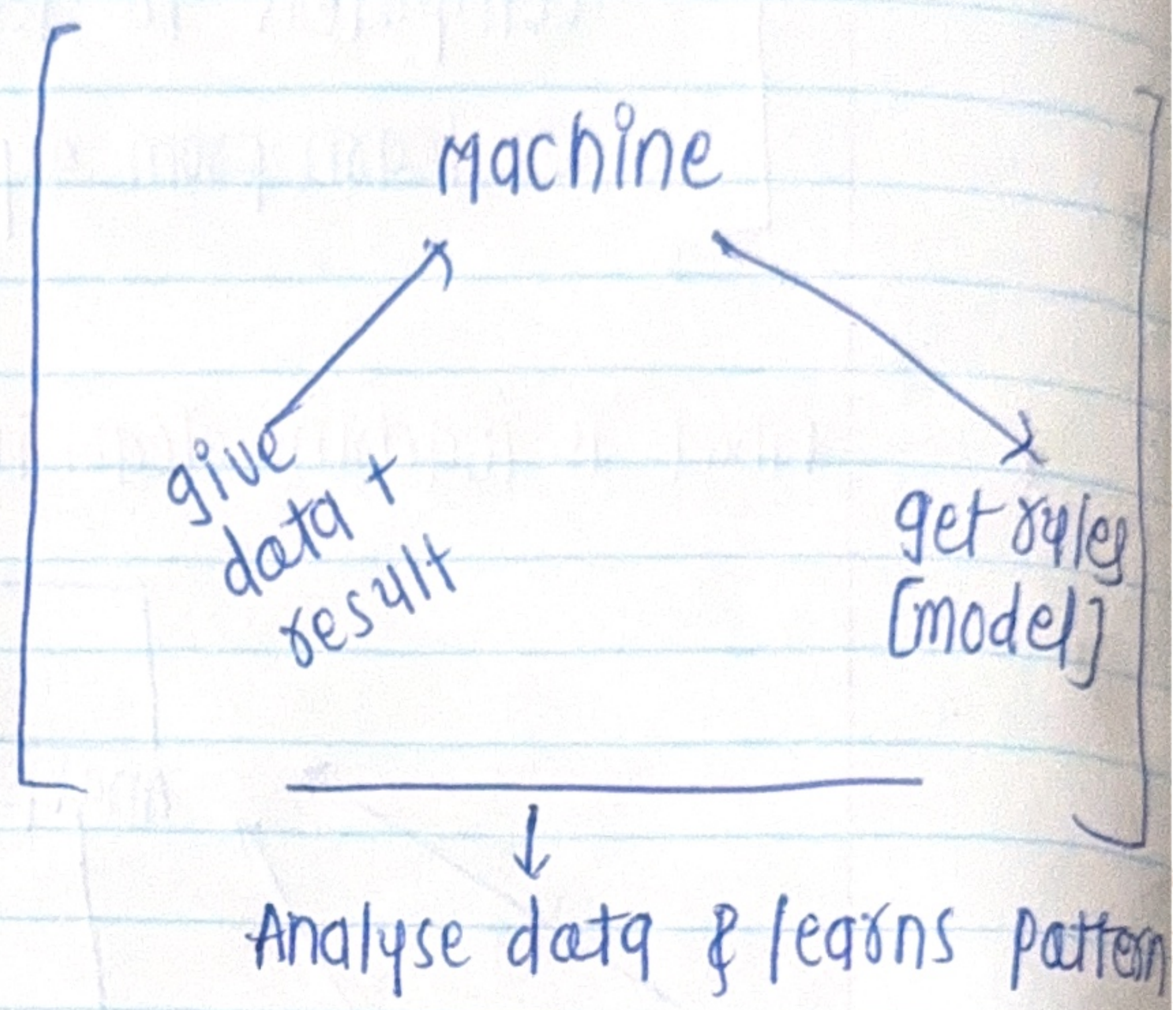



- Example:

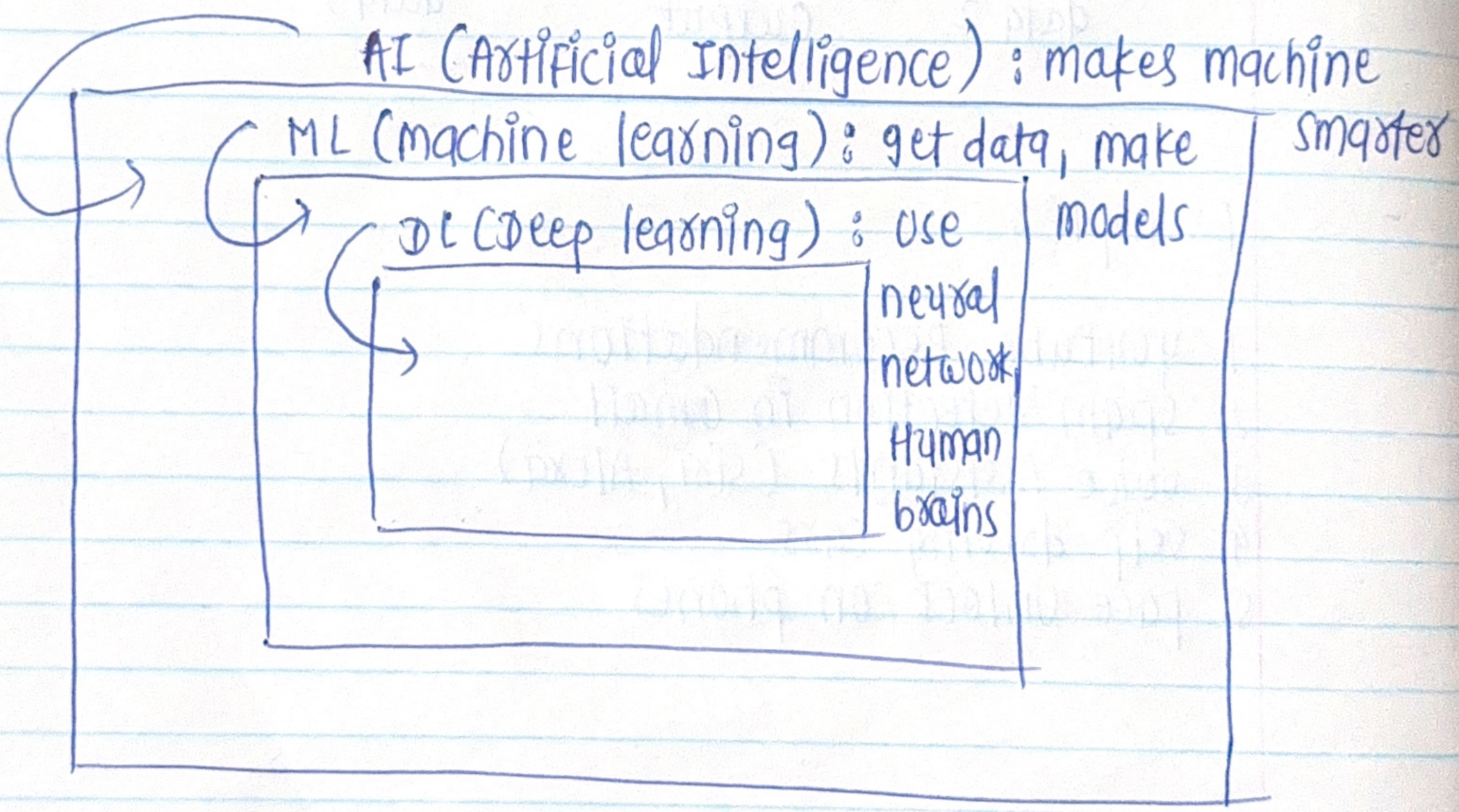
1. youtube Recommendations
2. spam detection in Gmail
3. voice Assistants (siri, Alexa)
4. self-driving cars
5. face unlock on phones

* Difference between traditional programming & machine learning :




* AI VS ML VS DL :



## * Machine learning Types:

1. Supervised learning
→ like student being taught by a teacher.
→ give input, result
↓
get model (learns prediction)

2. Unsupervised learning
→ get data but do not predict. Just find patterns.
→ Example: Google search

3. Reinforcement
→ like training a dog.
→ reward good behaviour, ignore/punish bad.
→ trial & Error.

## * Steps of making a machine learning model:

1. problem definition
2. Data collection
3. Exploratory data analysis (EDA)
4. Data processing /cleaning
5. feature selection & Engineering
6. Split the dataset
7. Model selection
8. Model training
9. Model Evaluation
10. Hyperparameter Tuning
11. Model Testing / v